

Contribution of aural cues to refine user spatial mapping in a virtual environment

Introduction and background

In recent years, virtual interior tours have become increasingly more popular as an integration into real estate services. The benefits of virtual touring of a space include geographical accessibility and convenience, for people who cannot travel to the physical location for touring or wish to forgo it due to inconvenience.

However, an issue with purely virtual interior tours is that human depth perception in virtual spaces tends to be inaccurate; this includes VR. Depth perception is specifically underestimated in VR (Jones et al., 2008) and the way the human accommodation system responds to portrayal of three-dimensional space in a two-dimensional window causes people to make spatial errors while representing depth in VR. (Wann et al., 1995)

In the context of clients using virtual interior tours to determine for themselves if a particular space is suitable for their habitation, being able to produce an accurate mental spatial mapping is important for the client to properly evaluate the feeling that each unique interior space evokes for them. Existing research demonstrates that different interior space designs evoke varying mental and physiological responses specific to the space. The perceptible physical properties and visual accessibility of a space which is functionally effective can still render it to be psychologically unsupportive, and there is discrimination between these and environments considered to be psychologically supportive. (Ulrich, 1991) The difference in interior visual surroundings also evokes different physiological responses in humans (Tsunetsugu et al., 2005) In the practical application of this problem, we would like to mitigate the spatial percept error of VR in order to provide clients with the maximum capacity possible to evaluate their individual level of comfort in a given interior space.

We can generalize the aforementioned application more to make it applicable to a wider variety of applications. The situation of a virtual interior tour for real estate purposes can be reframed into an expert-guided tour of a virtual space to a novice. Here we operationalize the definitions of “expert” as someone who is well-acquainted with the physical space in real life, and of “novice” as someone who has never seen the physical space and will be seeing it for the first time in the form of a virtual environment, such as VR. This reframed issue can be applied to other scenarios - a landlord remotely assisting the tenant of an unfurnished mixed-use space (e.g. for retail or teaching) to understand a possibly spatially complex interior can allow the tenant to successfully predetermine the optimal arrangement of furnishings before their arrival.

Research questions and underlying motivation

Findings in research have implied that hearing and other senses are effective as compensatory cues to improve object memory and sense of presence in virtual environments (Dinh et al., 1999), and we can use this knowledge to supplement the deficiency of rich visual information available in virtual reality. The anchor-point hypothesis for spatial cognition asserts that humans create spatial cognitive maps by networking together nodes (reference points), or 'regions', in their environment. (Couclelis et al., 1987) Hence, the more salient the available cues for establishing reference points, the more robust and accurate the resulting structure of the cognitive map. Other studies have also found that the use of spatial audio in virtual reality enhances auditory source localization as well as subsequent spatial mapping for individuals with visual impairment (Picinali et al., 2014) and that a combination of visual and spatial auditory cues are more effective for target acquisition in VR. (Flanagan et al., 1998)(Pierno et al., 2005)

Studying the effects of using auditory cues to supplement the information a novice has available to create a spatial mapping of a brand new virtual environment (the issue mentioned in the previous section) could impart useful information about how aspects of virtual reality can be enhanced to improve the quality of spatial mapping created by the user. This has useful applications that are mentioned in the introduction. A testable hypothesis follows: we would like to see if the use of auditory cues when the expert is referencing points in the virtual environment (while touring to show the novice) will refine novices' establishment of reference points and aid them in creating a more accurate spatial map of the virtual space.

Approach

A first thought is to use an experimental user study to determine if aural reference points will really refine the accuracy of a user's spatial map in a space-touring scenario. The activity will be as follows: the participant (novice) is led by an experimenter (expert) on an audio-visually guided tour of a specific interior space. The nature of the space (being the actual physical space, or a virtual reality construction of the exact same space) will be the manipulated variable. Post-tour, the participant will be provided with a paper grid of the same dimensions as the space and asked to draw out the locations of various objects in the space.

There will actually be three conditions. The control will be the physical space, and two other conditions will involve a virtual reality construction of the physical space. One of the VR conditions will include spatially-specific audio, meaning that the expert uses spatial auditory cues in the user's experience corresponding to specific points they talk about in the space being toured. The other VR condition has audio which is not spatially-specific, in the sense that the expert will still perform the tour in the same way but the audio will be ego-centered with respect to the participant and not specific to areas of the virtual space.

The objective will be to determine through these conditions if VR with spatially-specific audio will provide a more accurate mapping than VR with non spatially-specific

audio, as well as comparing the accuracy of VR participant-produced mappings to those who were in the physical space condition.

Measurement will be performed through comparing distance ratios from the participant-illustrated maps to distance ratios in a “correct” map of the original space.

Cited Works
(listed in order of reference)

Jones, J. Adam, J. Edward Swan II, Gurjot Singh, Eric Kolstad, and Stephen R. Ellis. "The effects of virtual reality, augmented reality, and motion parallax on egocentric depth perception." In Proceedings of the 5th symposium on Applied perception in graphics and visualization, pp. 9-14. ACM, 2008.

Wann, John P., Simon Rushton, and Mark Mon-Williams. "Natural problems for stereoscopic depth perception in virtual environments." *Vision research* 35, no. 19 (1995): 2731-2736.

Ulrich, Roger S. "Effects of interior design on wellness: Theory and recent scientific research." *Journal of health care interior design* 3, no. 1 (1991): 97-109.

Tsunetsugu, Yuko, Yoshifumi Miyazaki, and Hiroshi Sato. "Visual effects of interior design in actual-size living rooms on physiological responses." *Building and Environment* 40, no. 10 (2005): 1341-1346.

Couclelis, Helen, Reginald G. Golledge, Nathan Gale, and Waldo Tobler. "Exploring the anchor-point hypothesis of spatial cognition." *Journal of Environmental Psychology* 7, no. 2 (1987): 99-122.

Picinali, Lorenzo, Amandine Afonso, Michel Denis, and Brian FG Katz. "Exploration of architectural spaces by blind people using auditory virtual reality for the construction of spatial knowledge." *International Journal of Human-Computer Studies* 72, no. 4 (2014): 393-407.

Flanagan, Patrick, Ken I. McAnally, Russell L. Martin, James W. Meehan, and Simon R. Oldfield. "Aurally and visually guided visual search in a virtual environment." *Human factors* 40, no. 3 (1998): 461-468.

Pierno, Andrea C., Andrea Caria, Scott Glover, and Umberto Castiello. "Effects of increasing visual load on aurally and visually guided target acquisition in a virtual environment." *Applied ergonomics* 36, no. 3 (2005): 335-343.

Dinh, Huong Q., Neff Walker, Larry F. Hodges, Chang Song, and Akira Kobayashi. "Evaluating the importance of multi-sensory input on memory and the sense of presence in virtual environments." In *Virtual Reality, 1999. Proceedings.*, IEEE, pp. 222-228. IEEE, 1999.